found that they were extraordinarily intensive. Even on the world's fastest computer, they would take more than a century to complete. We had made progress, but we clearly needed a new idea, one that could greatly enhance the efficiency of our calculational method. Unwittingly, Victor Batyrev, a mathematician from the University of Essen, revealed such an idea through a pair of papers released in the spring and summer of 1992, Batyrev had become very interested in mirror symmetry, especially in the wake of the success of Candelas and his collaborators in using it to solve the sphere-counting problem described at the end of Chapter 10.

With a mathematician's perspective, though, Batyrev was unsettled by the methods Plesser and I had invoked to find mirror pairs of Calabi-Yau spaces. Although our approach used tools familiar to string theorists, Batyrev later told me that our paper seemed to him to be "black magic." This reflects the large cultural divide between the disciplines of physics and mathematics, and as string theory blurs their borders, the vast differences in language, methods, and styles of each field become increasingly apparent. Physicists are more like avant-garde composers, willing to bend traditional rules and brush the edge of acceptability in the search for solutions. Mathematicians are more like classical composers, typically working within a much tighter framework, reluctant to go to the next step until all previous ones have been established with due rigor. Each approach has its advantages as well as drawbacks; each provides a unique outlet for creative discovery. Like modern and classical music, it's not that one approach is right and the other wrong—the methods one chooses to use are largely a matter of taste and training.

Batyrev set out to recast the construction of mirror manifolds in a more conventional mathematical framework, and he succeeded. Inspired by earlier work of Shi-Shyr Roan, a mathematician from Taiwan, he found a systematic mathematical procedure for producing pairs of Calabi-Yau spaces that are mirrors of one another. His construction reduces to the procedure Plesser and I had found in the examples we had considered, but offers a more general framework that is phrased in a manner more familiar to mathematicians.

The flip side is that Batyrev's papers invoked areas of mathematics that most physicists had never previously encountered. I, for example,